Bangladesh University of Engineering and Technology
Department of Computer Science and Engineering

Course: CSE 206
Digital Logic Design Sessional

Experiment No. 5

Topic: Designing Circuits using Adder IC 7483

Design and implement the following problems:

1. Suppose, you have to design a circuit which acts like a BCD code to Excess-3 code converter, if the controller bit is 0 and acts like an Excess-3 code to BCD code converter, if the controller bit is 1. Design and implement the circuit using Adder and XOR ICs.

Decimal Digit	Excees-3 Code	BCD Code
0	0011	0000
1	0100	0001
2	0101	0010
3	0110	0011
4	0111	0100
5	1000	0101
6	1001	0110
7	1010	0111
8	1011	1000
9	1100	1001

2. Design and implement the circuit for the following problem using XOR gates and one 7483 IC. You are not allowed to use ***inverter***. You can use the XOR gates for inverter functions if needed.

Input	**Output**
00	X+1
01	X
10	X'+1
11	X'

Here "+" means addition operation and X is a 4-bit number.

Answer the following question:

1. Design a circuit that finds 9's complement of a BCD digit if the controller bit is 0 and 10's complement of a BCD digit if the controller bit is 1. Use a 4 bit adder and other gates.

2. Using Adder IC 7483, design a 3-bit comparator circuit which will have three outputs indicating whether P=Q, P>Q, and P<Q. Here, P ($P_2P_1P_0$) and Q ($Q_2Q_1Q_0$) are both 3-bit numbers.

Report:

For each of the problems/questions report should cover:

- Problem definition.
- Required equation in minimized form with necessary steps and corresponding Truth table.
- Circuit diagram with pin number.
- Required instruments for implementation.

Moreover, include the following sections to your report.

- Answer to the questions.